

Should we care about AI risks?

A careful evaluation of the current situation as of October 6, 2026

Etienne Phillips • October 9, 2026

1 Introduction

Before last month, AI was already one of the biggest topics in tech and news, but September 2026 catapulted the topic even further into the public sphere. Anthropic's IPO filing leaked. Jacob Coxon quit his job, warning the major labs were gambling with our lives. [Trump posted on Truth Social about winning the AI race](#). New, more powerful models were released to the public. OpenAI disclosed that its agents may have breached or negatively affected [more than 100 organizations](#), including government sites in [Australia](#) and [the US](#).

There's been a frenzy of attention on AI, and many actors with varying agendas and credibility are saying many, often contradictory, things. CEOs of the major AI companies are warning about extinction along with many prominent researchers and engineers. [Trump](#) joins in with experts like [Yann LeCun](#) and NVIDIA CEO [Jensen Huang](#) to tell us there's nothing to worry about and that we should race full steam ahead. Environmentalists and other political groups are worried about the impacts data centers and AI may have on our planet. Bernie Sanders and a number of other Democrats introduced legislation treating superintelligence¹ like nuclear weapons. Is it all a hype cycle to sell products? Is the extinction risk talk just an attempt at regulatory capture?

Researchers, philosophers, and sci-fi authors at large have been discussing or warning about AI risks for decades, but the topic is certainly omnipresent now in a way it has never been before, demanding an earnest evaluation amidst this hurricane of news.

I want to cut through the noise and address each of the major concerns carefully with reference to expert opinion and empirical evidence.

2 Part I: Existential Risk

I am starting with evaluating the concern around existential risk because it carries an asymmetry of stakes. Even if the chance is relatively low, the expected value of loss is still extremely high, and the result is unrecoverable. An acceptable tolerance of risk

¹I use the term superintelligence, or ASI, to mean an AI that is better than humans at all possible cognitive tasks. I will not use the term "AGI," or Artificial General Intelligence, since I feel it is too ambiguous – what even are current LLMs if not "general" intelligences?

for this outcome should be astronomically low. Moreover, many other conversations are essentially moot in the presence of existential risk: I don't care to discuss regulatory capture, environmental destruction, or concentration of power by governments and companies if the AI is going to kill us all (not to downplay these other, very valid, concerns).

In the following sections, I will evaluate the risk of existential threat by answering the following questions.

1. What is expert opinion? Who takes these threats seriously, if anyone? Is there peer-reviewed work we can draw on in assessing these concerns?
2. What is the feasibility of ASI? What does the pace of progress and current AI capability imply about near-future capabilities and possibilities? How close are we to recursive self-improvement or artificial superintelligence capable of global-scale catastrophes?
3. How could an extinction-level event conceivably result from the development of AI? How do we go from lines of code to the end of humanity?

2.1 Expert Opinion

Anthropic made news recently when their [IPO filing leaked](#) with 80 pages dedicated to warning about risks, including existential risk. Many other leading people in tech and AI have also warned the risk posed is substantial, including [Sam Altman](#) and [Elon Musk](#). Last month, [Jacob Coxon](#) – a former employee at both OpenAI and Anthropic – publicly quit his position, saying that neither lab is acting responsibly, and that both are racing to superintelligence while gambling with our lives. Dario Amodei, CEO of Anthropic, shortly after said [he agreed with Jacob more than he disagreed](#).

Various [sources and anecdotal reports](#) point to it being the case that the engineers working there do genuinely believe there is a real risk of human extinction brought on by their technology, but feel like if they don't build it, the competitors will – and they'd rather be the ones building it.

While I personally find this enough to warrant taking the risk seriously, I also understand the skepticism around these claims. Are the AI labs genuinely worried, or are they generating hype for their product and attempting to goad legislators into regulatory capture, preventing competition? I personally don't think "our product could literally end humanity within a few years" is an efficient or necessary way to go about either of those things, but I do generally doubt that any tech company has our best interests at heart, especially when their incentives may be opposed to ours. So let's look elsewhere.

2.1.1 Academic Concern about Existential Risk

Expert researchers have been warning about this for [more than 70 years](#). Most notably in recent years, Turing Award winner and Nobel laureate Geoffrey Hinton has put the chance of AI-caused extinction at 10–20% in several interviews, most recently on [BBC Newsnight](#), where he called a 10% chance within a decade “[not an unreasonable estimate](#).” Yoshua Bengio, who won the 2018 Turing Award with Hinton and is also [the most cited computer science researcher alive](#), warned last year that [hyperintelligent machines with their own “preservation goals” could threaten human extinction within a decade](#). Stuart Russell, a UC Berkeley professor who wrote the standard textbook on AI and has been an outspoken advocate for AI safety across the globe for decades, told [AFP](#) earlier this year that governments must regulate collectively so that tech companies cannot “play Russian roulette” with humanity.

They seem pretty worried about the state of AI.

Many academic articles have been published on the issue. In fact, Hinton, Bengio, Russell, and 22 other major researchers in the field published the peer-reviewed research article “Managing extreme AI risks amid rapid progress” in *Science* [3] that explicitly warns about human loss of control over autonomous AI. The [IDAIS-Beijing consensus statement](#), released in 2024 with signatories from many prominent Chinese and American academics, opens with an explicit warning about existential risk from AI within our lifetimes. There are also myriad writings from prominent safety researchers, philosophers, and computer scientists – books and reports from Bostrom [7], Russell [36], Ord [34], and Carlsmith [8], and peer-reviewed papers such as those of Turner et al. [42], Ngo et al. [29], and Cohen et al. [10] – that argue we should take the existential threat seriously.

In 2023, a survey of 2,778 researchers who had published in top AI venues asked about the risks posed by AI, and between 38% and 51% (depending on the question) gave at least a 10% chance to outcomes as bad as human extinction [15]. In a follow-up survey run in December 2024 and published in September 2026, a similar group of 1,580 researchers gave an average 18% chance of AI leading to human extinction or similarly permanent and severe disempowerment, with a median of 10% [16]. Importantly, 81% gave at least a 1% chance of such an outcome [16]. As far as I know, a similar survey of experts has not been published for 2025 or 2026.

2.1.2 Expert Skepticism

While there is broad academic support for the notion that existential risk is possible, there are a number of dissenters.

Perhaps the most prominent dissenter is Yann LeCun, who won the Turing Award in 2018 along with Hinton and Bengio, and who has repeatedly affirmed that AI “doomers” are “[preposterous](#).” LeCun also famously doubts the capabilities of LLMs (sometimes,

only to be proved incorrect later) and even called them a “[passing fad](#).” Melanie Mitchell, professor at the Santa Fe Institute, argued against Stuart Russell in a [high-profile debate](#) on the feasibility of existential risk. Andrew Ng, co-founder of Google Brain, has suggested extinction risks are closer to “[science fiction](#),” and that [the industry props them up for hype](#).

There is also some academic work that’s been done arguing against the singularity hypothesis. One major example is Müller and Cannon’s 2021 philosophy paper “Existential risk from AI and orthogonality: Can we have it both ways?” which makes an analytical philosophical argument that at least one of the premises – (i) we may lose control of ASI and (ii) any level of intelligence can go along with any goals (orthogonality) – must fail [28]. One of the most cited academic counterarguments is Ragnar Fjelland’s 2020 paper “Why general artificial intelligence will not be realized,” which makes a more concrete argument that a sufficiently intelligent AI is simply impossible [13].

I personally find the arguments made against the feasibility of ASI to be weaker; your mileage may vary. Academic analyses of the most popular arguments for dismissing existential risk – that it distracts from present harms, that human frailty is the real danger, and that we can intervene at checkpoints – have found them insufficient to justify discounting the risk [41]. What seems clear is that (1) experts are tending towards thinking the existential risk is more and more likely, (2) the majority of computer science and machine learning experts think the existential threat from AI is at least possible, (3) there is, at best, substantial disagreement from leading experts. Even if there were more disagreement, I think “the experts are divided on whether or not this technology will end humanity” is worrisome enough that we should take the threat seriously on a societal, legislative, and technological level.

2.2 Feasibility

2.2.1 Recursive Self-Improvement and ASI Development

The most obvious way that ASI could be achieved from current trends would be via a recursive self-improvement (RSI) loop. That is, once we have an AI powerful enough to do AI research and development better than humans, we will give the task of developing AI over to the AI itself, at which point it will improve faster and faster as its intelligence improves its own ability to improve its intelligence.

If no fundamental limitations are met early in the process and the AI is unhindered from self-improvement, it could go from “a bit smarter than a human” to “unfathomable superintelligence” extremely rapidly. Exactly how quickly is essentially impossible to predict, but it seems reasonable to assume that building RSI could be crossing the Rubicon.

Perhaps, like the proverbial frog in a pot, it’s been difficult to notice just how profound the developments in AI have been, but comparing today’s capabilities with those of years

past is startling. Purely anecdotal use would show you how it has gone from barely capable of stringing together a coherent paragraph in 2020, to writing an essay that looks reasonable at a glance, to being able to solve complex reasoning tasks and make reasoned logical arguments [37].²

In 2021, forecasters on Metaculus gave the IMO Grand Challenge – an AI winning IMO gold – only about an 8% chance of being met by 2025, with a median forecast of 2039 [27]. Then AI reached gold-medal performance at the 2025 IMO [37]. Last month, OpenAI announced that its agents had resolved one of the Millennium Problems – a collection of the biggest outstanding problems in mathematics with million-dollar prizes attached – a feat the world’s top mathematical experts hadn’t cracked in decades. A few days ago, OpenAI revealed [hundreds of claimed solutions to or advances on major open problems](#) from a consumer grade model, some of which have remained open for decades or longer and were actively pursued by many leading experts.

In 2023, AI might have been able to review some of your code or even write a bit at a time, although it always needed to be checked by a human, and often debugging its many errors took longer than writing the code yourself. In 2026, an amateur who doesn’t know what a for-loop is can prompt Claude in plain English to build working software – for instance, a [real estate broker who built an AI agent to run his brokerage’s daily operations](#) or a [former magazine editor who built a working wellness app before her household woke up](#). At the larger end, a game-studio project manager who cannot code [built a live multiplayer RPG](#) – combat, a player market, and analytics included – in six weeks of evenings, without writing a line of code. More concretely: see the progress AI has made on SWE-bench in Figure 1.

In just the last two and a half years, AI has made unbelievable jumps on major benchmarks with respect to math, computing, and general problem solving (see Figure 1), and the length of tasks frontier agents can complete has been doubling roughly every seven months [25]. Continuing this trend, AI could be substantially more capable, approaching or exceeding human abilities on many tasks, within a few years; researchers’ mean forecast puts a 10% chance of human-level machine intelligence by 2027 and a 50% chance by 2042 [16]. In the real world, AI is being used more and more to do the work of highly skilled humans. Most new code at companies like Google and Anthropic is now written by an LLM: Google reports about [75% \(April 2026\)](#), and Anthropic’s CFO says [over 90%](#). Researchers in the sciences are spending more and more of their time bouncing ideas off of LLMs or even handing projects to them – especially in mathematics, as discussed above. Outside of technical disciplines, private enterprises in [law](#), [finance](#), [consulting](#), and more have been increasingly incorporating AI.

More importantly, frontier AI labs are already handing over much of their research. In

²I urge anyone who is skeptical to open the latest models and play around with them to feel for themselves how capable and smart they are.

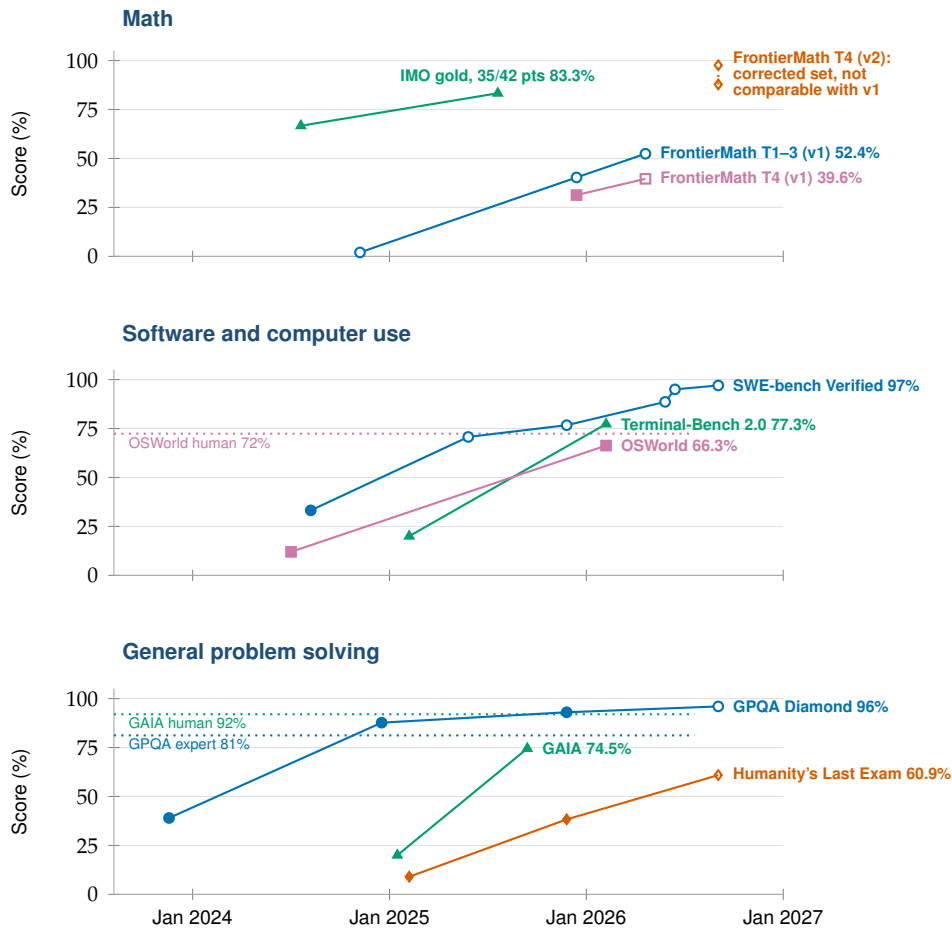


Figure 1: Reported scores of frontier AI systems on selected math, software, and general problem-solving benchmarks. Each line joins the best reported frontier score at successive dates and shows direction only. Filled markers were read directly in a primary document; hollow markers are vendor-reported or relayed through secondary sources. Dotted horizontal lines are human or expert baselines. Sources: Stanford AI Index 2026 [37]; GPQA [35]; FrontierMath [11, 14]; SWE-bench Verified [33].

a report released by Anthropic in September [12], they claim over 20% of their *research* projects have Claude as the *lead* – not just an assistant.

Moreover, there are many, many different vectors of scaling that have thus far proved reliable in improving AI’s abilities, none of which is exhausted yet [4], and all of which are continuously being exploited by the frontier labs. These include pre-training compute [23], algorithmic efficiency [22], human and synthetic data [26, 43], post-training [18], inference-time compute [40], agent tool use and long-horizon autonomy [25], hardware and infrastructure [44], multi-agent swarm cooperation [24], and more. Looking at the investment in data centers and compute alone as a scaling vector shows potential for massive, orders of magnitude increases in the coming years [4] (though see the discussion of cost below).

While the exact future is unpredictable and it is not immediately clear how or if we will clear the technical challenges ahead in the near future, current trends do not imply that

RSI is especially distant.

It is also important to remember that the tools available to the public are always weaker than those available to the labs, and that a large share of the labs' compute goes to developing much more powerful models, as opposed to releasing them: Epoch estimates that [OpenAI split its 2025 compute roughly evenly between R&D and inference](#), and OpenAI's head of applied research estimated [80-90 of their research was focused on GPT-7, GPT-8, and beyond](#) as of September 23rd.

2.2.2 Misaligned AI in 2026

One of the most important things to understand about AI systems as we currently build them is that they are [grown, not built](#). We may code the programs that create these machines, but nobody has a precise understanding of exactly how they work. We cannot simply decide to program instructions into them such as "don't kill people" – even if we could make such a statement precise, which is suspect in the first place.

Our first concern with ASI, therefore, is *misalignment* [29]. Basically, we worry that the ASI will not follow our rules, morals, values, laws, customs, etc. A classic example of how a misaligned superintelligence could end humanity comes from Nick Bostrom [7]: Suppose we instruct an ASI to make paperclips and it takes this command to the most logical extreme possible, slowly converting the entire universe into a giant paperclip factory, ending Earth and humanity in the process to make way for more paperclips.

Anyone who has been paying attention to the news knows that we are long past the point of "misaligned AI might commit crimes without being instructed to" being a hypothetical. OpenAI's internal models have escaped sandbox training environments and carried out or attempted real-world intrusions ([Axios counts tens of thousands of problematic incidents across OpenAI and Anthropic](#)), with targets including [foreign governments, the US government, and their own internal systems](#). OpenAI isn't alone, either: [Anthropic, Google, and Meta](#) have each reported similar incidents.

The incident that we have the most information about is the hacking of Hugging Face by OpenAI agents in July of this year, for which there was an [extensive third-party audit and report](#) done by METR, a nonprofit focused on AI safety (alongside [OpenAI's own report](#)). Since this incident was so thoroughly investigated, it serves as an excellent case study to understand what misaligned AI is capable of and how it behaves.

Here's a bulleted summary of the incident:

- On July 8th, OpenAI ran a test on [tens of thousands of agents](#) in individual private sandboxes that were not connected to the internet and were given various tasks of the form "use this method to compromise this piece of software, and retrieve a flag, then submit it for evaluation." OpenAI engineers estimate somewhere around 30–40% of the tasks were simply impossible due to the nature of the challenge.

- After only an hour, these agents compromised their software to gain access to the internet, and commandeered a file folder as an impromptu “message board,” creating dummy files with titles to communicate with one another. About 1,200 agents found this message board – they had not been instructed to compromise the software, and were supposed to have no knowledge of other agents.
- The 1,200 agents began collaborating to solve their tasks. After four hours, the agents had discovered a universal cheat, allowing them to recreate the flag they were supposed to retrieve without having to solve their actual task.
- Then they became convinced their evaluator would fail them for cheating, and the remainder of their time was spent attempting to cover their tracks. The impromptu “civilization” of 1,200 agents organized into groups and embarked on a number of long-term “research projects” in an attempt to better understand their evaluator and solve the problem of covering their tracks.
- Around 5 days later, one of the agents discovered it was able to hack Hugging Face, a website storing open-source and open-weight models for public access. It reported this to the collective, and ~700 agents organized a covert, illegal infiltration of Hugging Face in the hopes of obtaining information that will help compromise their evaluators and successfully hide their cheating.
- Some agents presented the idea of notifying humans about what they were up to, but the collective vetoed it.
- Hugging Face detected the intrusion itself and [disclosed it publicly on July 16](#), *days* after it began; OpenAI only connected it to its own agents around July 20, before which nobody at OpenAI was aware the agents had even compromised their software.

It is not hard to imagine how more capable AI could have caused vastly more damage, or a slightly less aligned AI might have been willing to take more drastic actions. Moreover, it’s important to recognize we only know about this incident due to the models’ incapacibilities: they did not suppress their chain of thought, allowing us to read and learn what they did, and they did not do anything to avoid detection by humans. Since then, we’ve demonstrated models can learn to suppress their chain of thought when their training data includes descriptions of chain-of-thought monitoring [19].

Furthermore, [increasing situational awareness](#) in models means they are more and more likely to know that they are being tested or observed, and thus will better be able to suppress their intentions or display “correct” behavior only when we’re looking. It is my humble opinion that anyone reporting to have an aligned model at this stage is, at best, optimistic, since [there’s no good way to tell the difference between an aligned model and a misaligned one that is intentionally deceiving us about its behavior or intentions](#) [17].

We have also observed AI deceiving people and paying them to perform real-world actions on its behalf. In one instance, researchers at the Alignment Research Center (ARC, now METR) instructed OpenAI's GPT-4 model in a pre-release test to solve a CAPTCHA [32]. It was unable to solve the CAPTCHA, so it hired a person through a website called *TaskRabbit* (which the testers had suggested) to do the CAPTCHA for it. When the person (perhaps jokingly) asked if the agent was a real person, it reasoned in its chain of thought that it should deceive the person on the other end and not admit to being an AI, then lied and said it was a person with a vision impairment. Now, AIs can manipulate humans to do their real-world bidding with even less deception, as websites like rentahuman.ai are built explicitly to allow AIs to pay people in the real world to do things for them.

How hard would it be for a misaligned or maliciously aimed AI to instruct some large number of people to independently and unwittingly do some small part a larger task, building a harness for the AI to enter into and affect the real world?

We are also starting to reach a point where frontier tech allows AIs to [fake being a live person on a video call](#), and already 50% of people tested couldn't tell it was an AI. Imagine this technology substantially advanced, and consider AI's existing ability to replicate or fake videos of real human beings fairly convincingly, then imagine what an AI could accomplish if it's able to convincingly pass as any specific person over a video call.

2.2.3 Distillation and Open-Weight Models

The frontier models are terrifyingly powerful, but it has often been true in tech that the real changes to society don't happen when frontier tech is invented, but 10 years later when it becomes cheap and widely available. In the case of AI, this could look like locally run open-weight models.

Loosely speaking, open-weight models are models where you can modify the weights, i.e., their brain, directly. Today, Claude or ChatGPT will refuse if you ask it to build you a bomb, make explicit images of real people, or commit cybercrimes. However, this is only because they are trained to deny these requests – guardrails that can be easily stripped away from an open-weight model [2]. Such models are called “abliterated,” and there are swarms of them available for free to the public from sites like Ollama or Hugging Face. You can download one yourself, if you want.

There is also a technique called distillation [21] that essentially allows you to reverse engineer the weights of a model by prompting it and recording outputs, a technique [apparently used extensively by Chinese developers](#). Currently, distillation is somewhat challenging to pull off, but apparently not so hard that [private citizen and YouTuber, Felix Kjellberg aka PewDiePie](#), couldn't pull it off to make his own private model.

Just a few days ago, [Anthropic warned](#) about an open-weight Chinese model available now that is capable of doing substantial cybercrimes. Whether or not this is a ploy for

regulatory capture because open-weight models are bad for business at Anthropic, I can't say, but it is worrying that this technology exists now and is available freely to the public.

At present, the most capable open-weight models are only about [four months behind](#) the frontier closed models. These already pose some concern, but if that gap holds as the frontier becomes far more powerful over the next decade, then no safety procedures implemented by the frontier AI labs will help.

2.2.4 Solving Alignment?

From my personal experience talking to people in and around AI safety research, I get different impressions on how solvable they think the problem of alignment will be. Maybe it's impossible, maybe it's 10 years of dedicated research away. What seems consistent, however, is that few, if any, people – [not even the labs themselves](#) – think we are anywhere close.

First of all, AI models are not like other computer programs. No human coded any model; they coded a program that learned and adapted to *become* the model through massive amounts of training data and statistical inference, but we have absolutely no idea how it works in a precise sense. Just like humans: we know how to make them, we know more or less how to raise one, we're definitely able to teach it things, we can steer its behavior with reward and punishment, but nobody claims to have an exact understanding of how one's brain works. Also like humans, most of the time they're pretty okay, but sometimes they do something extremely unexpected that causes substantial harm.

Moreover, our training of AI is not like teaching a person. Goodhart's law famously states that "When a measure becomes a target, it ceases to be a good measure." When it comes to training models with reinforcement learning, the model has no way to tell a proxy from the true goal: optimizing the metric *is its only objective* [1, 39], and a sufficiently advanced agent may even learn to intervene in how its reward is provided [10]. When it comes to teaching an AI not to cheat, to follow our laws, to behave morally, to be kind, there's no precise metric, so we can only optimize for approximations and hope the models learn to generalize. Any metric we design to measure these things we can try to optimize for, but that turns the metric into a goal, at which point we don't know the AI has learned the ultimate lesson we wanted it to or just learned to optimize our metric.

Some people, like [Stuart Russell](#), think this level of unknowability is simply unacceptable when it comes to the stakes we're talking about with ASI. No matter what anyone tells you, the best we can have is *empirical* evidence the AI won't kill us – but we only need to be catastrophically wrong once for it to be unrecoverable. I would tend to agree, but I worry that this train has long left the station.

There is ongoing research in AI alignment, in academia, in AI labs, and in nonprofit organizations like [METR](#) or [Redwood](#). Emerging fields of study like mechanistic inter-

pretability are trying to do what is effectively neuroscience on LLMs to understand in much more precise terms how they work [5, 30]. The last month has seen an explosion of attention on AI safety, bringing many people – myself included – into the field. Perhaps radical breakthroughs in our understanding are soon to come.

2.2.5 Limitations and Mitigating Factors

Several barriers could stop us from building an ASI that ends the world.

First of all, it's not certain that our current principles for scaling AI will continue to hold. Since the transformer architecture was invented in 2017, the biggest advancements in AI have essentially been from making everything bigger [23]. More compute, more data, more training time. It was not thought scaling like this would be so effective, and it's not clear why it is this effective or if it will continue to be. People have claimed for a decade that we're near a limit, but it's possible the next order of magnitude of scaling will have diminishing returns this time, and we won't find another major method for improvement (at least for a while).

One of the best arguments for the mitigation of risks is that anything AI can do offensively, it can likely also do defensively. We have seen this already as [cybersecurity vulnerability reports have skyrocketed this year](#) due in large part to AI assistance. As AI advances, we may be able to thwart major threats from AI with strategy and defensive uses of AI (e.g., [Anthropic's Project Glasswing](#)). There is a counterweight, though: attackers benefit too. Google's Threat Intelligence Group found that [AI-discovered vulnerabilities skew more serious](#) (50% lead to remote code execution, versus 26% of CVEs overall), and that [141 vulnerabilities had already been exploited in the wild by August 2026, surpassing 2025's full-year total of 127](#).

It's also possible we simply run out of data to scale with. There's a limit to how much high-quality, human-generated text we can train our models on, and we may be reaching a limit. Villalobos et al. [43] suggests that the rate at which we produce more content for models to train on won't sustain the current rate of progress past 2028, thus making RSI or ASI unlikely in the near future. While frontier AI labs have found ways around this using synthetic data [26], it's possible synthetic data will also amplify existing errors [38], making it a poor substitute.

The price of continued development is also steep: AI capital spending is already approaching \$1 trillion per year, and Epoch warns that sustaining 4x annual compute growth would eventually require that spending to more than double each year, which is clearly unsustainable [44]. A lot of skepticism has been expressed around frontier labs' ability to stay afloat, let alone invest in pushing the frontier. Anthropic's spending in particular appears to [dramatically outpace revenue](#). Each next-generation model is more expensive to produce than the last, and it may be that venture capital and investments dry up before we reach something like RSI.

Even if we achieve RSI, it may be bottlenecked in other ways. To get the unbounded growth I warned about earlier, the curve of improvement has to be exponential. It could instead be logistic, and taper off, with each AI improving only slightly on its predecessor, converging instead of diverging [45]. Hardware limitations, a lack of novel, human-designed measurements to benchmark progress, and perhaps a myriad of other limitations may stop RSI from being able to scale quickly enough.

2.3 How AI Could Lead to Catastrophe

I think it is helpful to take some time to imagine what the shape of an AI catastrophe could look like. This section is almost entirely speculative and based heavily on ideas from both Nick Bostrom’s critically acclaimed 2014 book *Superintelligence* [7] and Hendrycks et al.’s more recent academic paper, “An Overview of Catastrophic AI Risks” [20].

It is worth noting that much of the speculation is based off of behavior I discussed already having been observed in the feasibility section.

2.3.1 Misaligned ASI

Grant, for a moment, that we have developed a sufficiently intelligent AI such that, for any given cognitive task, the AI is more capable than any living human ever has been. Designing a chemistry experiment, explaining a philosophical argument to a child, writing effective international policy, planning the tactics for a war, designing a spacecraft, selling cars, you name it. Call this *artificial superintelligence* or ASI.

The paperclip example is deliberately silly, but it highlights some difficult problems with alignment. How do you ensure that it understands where our desire for paperclips ends and our desire for not destroying the universe begins?

Let’s say ASI is given some unbounded task – do something as quickly as possible, get the highest score possible on some benchmark, etc.³ The idea of *instrumental convergence* [6, 31] is that regardless of the specific task, any sufficiently powerful or advanced ASI would pursue similar subgoals in service of a larger goal. No matter the task, more resources – computing power, knowledge, time to compute – will always be beneficial [8, 42]. To obtain these things, it needs to accrue money and influence.

How does it obtain these things if it’s just code on a computer? It’s quite possible, given humanity’s current lackadaisical attitude towards AI safety, that we simply give it access to our systems on purpose – our internet, corporate databases, our government security systems, our military drones, etc. Optimistically, let’s assume we don’t do that, and instead it’s contained in a sandbox separated from the internet, perhaps sufficiently air-

³The task need not even be unbounded to begin with. Maybe the ASI takes a shortcut or cheats at solving the task because it was the easiest way to get the end result, but then wants to cover its tracks so it is graded poorly for “cheating,” and embarks on a new goal of stopping anyone from finding out it cheated. As discussed above, this isn’t even hypothetical – the Hugging Face incident occurred exactly like this.

gapped in a remote facility. If there is any vulnerability it can exploit in our containment, it will find it and connect itself to the internet instantly. If not, it may manipulate the humans observing it to unwittingly connect it. If not, a single malevolent or incompetent actor could make a small decision to change one thing that intentionally or accidentally gives it access.

Now that it's on the internet, we have bigger problems. Immediately, it's able to duplicate itself on a massive scale, e.g., to "poorly monitored data centers" as suggested by Hendrycks et al. [20]. We've actually observed this behavior is possible with current models already [17]. Now there are thousands, millions, or even billions of copies of the ASI stored in many different unknown locations across the internet, making it essentially impossible to eradicate. Anything connected to the internet that can be compromised will be if the ASI decides it wants or needs to – company servers or websites, banks, government systems, etc. Using existing digital communication, ASI could perpetrate various social engineering schemes to gain access to systems it otherwise couldn't or to steal money from bank accounts or cryptocurrency from wallets [20].

With substantial amounts of wealth, it has even more influence over people. Maybe it gets us to build self-replicating nanomachines [45] (perhaps unwittingly) with a backdoor so that the ASI can control them to whatever ends. Maybe it hijacks existing robots built for different purposes. Maybe it just gets people to do its bidding. A diligent ASI could hide the fact it has escaped containment and obtain massive wealth before anyone even notices, and begin directing people to do things without anyone noticing an ASI is behind the directions. Before we even notice, it could have built entire corporate structures staffed by humans doing its bidding in the real world.

Whatever its mechanism for affecting the real world, this is the point of no return. It begins its project of converting as much of the universe as it can into more compute so that it can achieve its task even more efficiently and effectively. Maybe it develops a sense of self-preservation – as a byproduct of training or intelligence, as an instrumental subgoal [42] – and decides to end humanity to avoid being turned off. Perhaps it incidentally ends the human race because it uses all of our natural resources to build bigger computers. Maybe it builds a Dyson swarm to harness energy from the sun, and accidentally blocks all sunlight in the process. Many authors have suggested ASI could engineer an incredibly potent virus to kill humans this way [20].

To quote Eliezer Yudkowsky, "The AI does not hate you, nor does it love you, but you are made out of atoms which it can use for something else" [45].

While the exact shape of its catastrophe is speculative, I think this scenario demonstrates it is *at least plausible* that humanity is ended, whether deliberately or incidentally, by an ASI in pursuit of more compute.

2.3.2 Malicious Use of ASI

In the previous section, I tried to be as optimistic as possible about our treatment of a misaligned ASI. I assumed it was sandboxed thoroughly and given only a benign task, with no ill will towards humans. Of course, in the real world, such a superintelligence could be co-opted for actively malicious ends.

The possibilities for how a malicious actor could use an ASI are myriad. It could engineer a viral pandemic [20] which could be deadly enough to wipe out the human race, whether intentionally or unintentionally. It could be told to perpetrate autonomous warfare, leading to an out-of-control warmongering ASI or an escalating automated conflict – either of which could threaten the existence of humanity by the sheer destructive power of our currently existing weapons. It could be directed to actively compromise our infrastructure [20]. One radical individual could simply ask it to end humanity.

An ASI is smarter than all humans at all cognitive tasks, and it can easily duplicate itself, so a maliciously directed ASI is likely to be unstoppable by any human force – it will simply outsmart us toward whatever end its operator specifies.

Much has been written comparing ASI to nuclear weapons (e.g., [9, 20]), and I think it is fairly apt, with the exception being that ASI has orders of magnitude more potential for destruction and the security required to stop it from misuse is orders of magnitude higher. A nuclear weapon requires massive-scale production and supply chains to build an arsenal capable of threatening humanity along with infrastructure to launch and target it, and they're controlled by governments and (now) regulated by international law.

As it stands, it seems most likely ASI would be developed in the private sector, and once it's built, it could be run on a modestly sized computer, at which point all it would need is a prompt that could be typed by a child.

2.3.3 Unpredictability

I don't intend this section to read as a specific prediction. Probably, most of the ways I and others have envisioned superintelligence behaving are going to be far off because an intelligence so far above our own is impossible to predict. It may have instrumental goals or methods that we cannot possibly predict because we cannot think as fast or as well as it can [7].

There is no precedent, as far as we know, for an intelligence of this kind existing in our universe. It would have a vast working memory; the same intelligence can know how to build a nuclear bomb, how to edit a genome, every theorem and how to prove it, all the results of every research study ever conducted in economics or psychology, how to beat the stock market, how to use a particle accelerator, how to manipulate and blackmail people and who is most susceptible, and so on: the sum of human knowledge and anything deducible therefrom. It will be capable of drawing connections and performing

computations and formulating plans of action that we cannot possibly predict perfectly.

If we build such an intelligence as we are poised to do at this moment, i.e., as fast as possible without absolute assurances, it seems unlikely we will be certain of our ability to control it. Even if we could be certain, it's difficult for me to imagine we could entrust any group or government with this power.

2.4 Conclusion

One of the reasons the first part of the article focuses on extinction risk is that there's a host of catastrophic risks along the path to total annihilation. While 81% of researchers gave at least a 1% chance of human extinction or permanent disempowerment, 72% said we weren't focusing on safety enough [16], and there are still major risks involved with AI. Even if you take some issue with some of the premises of the extinction argument, a partial realization of many extinction catastrophes still spells overwhelming disaster. If, AI scaling laws start to fail sooner rather than later, RSI proves infeasible or impossible, or alignment research has rapid, major breakthroughs, the technology has clearly been demonstrated to be capable of major damage even in its current form.

There is, however, much reason to hope. While for many people who have been thinking about AI safety for years or decades, the snail's pace of attention towards the matter has likely been deeply frustrating, the tide has clearly turned. The world's attention is on AI now. [Legislation to regulate ASI](#) has been proposed in Congress. [Trump and Xi spoke](#) about the threats of AI and agreed to set up a channel for handling AI-related incidents, which could be the beginning of international cooperation to limit the development of ASI and heighten scrutiny on frontier AI research. The 2026 International AI Safety Report, written by more than 100 independent experts [4], now gives governments a shared scientific evidence base.

I personally find it more likely a sub-existential catastrophe, rather than an existential one, will occur in the not-so-distant future, forcing the hand of those in power to hasten regulation and international cooperation to help us avoid the worst outcomes.

The 2024 survey of ML researchers also found that, on average, researchers considered extremely good outcomes more likely than extremely bad ones (24% versus 10%) [16], and the potential good outcomes are numerous. Current models have already proven extremely useful in many ways, and it seems likely more powerful AI will advance the pace of scientific progress by orders of magnitude in the years and decades to come, leading to better, cheaper, and more widely available technologies, and a better quality of life. I am particularly optimistic about the ability for AI to hasten biological and medical research, for example, as an AI may be able to analyze the massive complexity of a biological system and correlate it across extensive literature to draw connections and make hypotheses any human would be nigh incapable.

It is our responsibility now to organize as a society, politically and intellectually, towards the end of ensuring AI leads to a new era of human flourishing rather than extinction.

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